LESSON 7.9

Reflect Points Across Axes

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can reflect points across the x-axis and y-axis on the coordinate plane.

WHY THIS MATTERS Captain Vega discovers that her treasure map has a magic mirror trick! When she folds the map along the x-axis, some clues match up perfectly with clues on the other side. For example, a marker at (2, 4) has a matching marker at (2, -4) when reflected over the x-axis. Another pair reflects over the y-axis: (5, 3) and (-5, 3). How does reflection change the coordinates?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Reflection *Spanish: Reflexión*	A flipped, mirror image across a line.	(3, 2) reflected over the y-axis becomes (-3, 2) — x changes sign, y stays
x-axis *Spanish: Eje x*	The line that goes across the grid. Flipping over it changes the y sign.	(4, 3) → (4, -3): the y flips from +3 to -3, like folding the paper along the horizontal line
y-axis *Spanish: Eje y*	The line that goes up the grid. Flipping over it changes the x sign.	(4, 3) → (-4, 3): the x flips from +4 to -4, like folding the paper along the vertical line
Symmetry *Spanish: Simetría*	When a shape looks the same on both sides of a line.	A butterfly's wings are symmetric — fold it down the middle and both sides match
Integer *Spanish: Número entero*	Whole numbers and their opposites, like -2, -1, 0, 1, 2.	..., -3, -2, -1, 0, 1, 2, 3, ...
Perpendicular *Spanish: Perpendicular*	Two lines that meet to make a square corner (90 degrees).	The dashed height meets the base at a square corner — the small 90° box is what makes it perpendicular, not slanted.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I will reflect (3, 2) over the y-axis.

1. The y-axis is the vertical line, so the point flips left or right. That changes the x-coordinate.
2. The x changes sign: 3 becomes -3. The y stays the same: 2 stays 2.
3. So the reflection of (3, 2) over the y-axis is (-3, 2).

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's reflect (4, -5) over the x-axis. Which coordinate changes over the x-axis? The y-coordinate.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- When I reflect over the x-axis, the _____ -coordinate changes sign and the _____ -coordinate stays the same. When I reflect over the y-axis, the _____ -coordinate changes sign and the _____ -coordinate stays the same.

Word bank Reflection | x-axis | y-axis | Symmetry | Integer | Perpendicular | reflect | opposite | sign | change | same | mirror

WATCH OUT!

A common mistake

A common mistake in Reflect Points Across Axes is flipping the sign of the wrong coordinate

LESSON 7.9

Reflect Points Across Axes

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each point and axis to its reflection.

_____ (2, 5) over y-axis	(A) (-2, 5)
_____ (-3, 1) over y-axis	(B) (3, 1)
_____ (4, -6) over x-axis	(C) (4, 6)
_____ (-1, -4) over x-axis	(D) (-1, 4)
_____ (5, 2) over x-axis	(E) (5, -2)
_____ (-6, 3) over y-axis	(F) (6, 3)

STEPS TO TRY

1. Read the axis on each card: the y-axis flips the x sign; the x-axis flips the y sign. Only ONE number changes.
2. For each card, copy the pair, flip the correct coordinate's sign, and find that result among the matches.
3. Try '(2, 5) over y-axis': flip the 2's sign and keep the 5. Find the match that shows exactly that change.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

What is the reflection of (4, -3) over the x-axis?

(A) (-4, -3)

(B) (-4, 3)

(C) (3, -4)

(D) (4, 3)

STEPS TO TRY

1. Reflecting over the x-axis flips the point UP/DOWN — the left/right position doesn't move.
2. Keep the x-coordinate the same and switch the sign of the y-coordinate. Don't swap the numbers' order.
3. The point sits 3 units below the x-axis. Its mirror sits the same 3 units above, with the x still 4.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is the reflection of (-2, 6) over the y-axis?

(A) (-2, -6)

(B) (2, 6)

(C) (2, -6)

(D) (6, -2)

STEPS TO TRY

1. Reflecting over the y-axis flips LEFT/RIGHT — the height stays the same.
2. Switch the sign of the x-coordinate and keep the y-coordinate 6 unchanged.
3. The point is 2 units left of the y-axis. Its mirror lands 2 units to the RIGHT, at the same height.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which point is on the y-axis?

(A) (-2, 4)

(B) (0, 5)

(C) (3, 3)

(D) (5, 0)

STEPS TO TRY

1. Points on the y-axis have gone 0 units left or right — check the FIRST coordinate.
2. Scan each pair's x-coordinate (the first number). Only x = 0 keeps a point on the y-axis.
3. Read the first number in each pair. A point sits on the y-axis only when that number is 0.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What pattern do you notice about the coordinates when reflecting over the x-axis versus the y-axis?

SENTENCE STARTER When I reflect over the x-axis, the ___ -coordinate changes sign and the ___ -coordinate stays the same. When I reflect over the y-axis, the ___ -coordinate changes sign and the ___ -coordinate stays the same.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help